

BIOSTATR: Applied Biostatistics with R

Full Program, September 2026 Cohort

Imad EL BADISY, PhD

1 Course Overview

Title: Applied Biostatistics with R

Format: Online, cohort-based

Duration: 2 weeks, 8 sessions, 2 hours per session, 16 contact hours

Dates: 07/09/2026 – 20/09/2026

Days: Monday, Wednesday, Friday (18h00 to 20h00), Sunday (10h00 to 12h00)

Time: Morocco time

Instructor: Imad EL BADISY, PhD

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1.1 Course Description

BIOSTATR provides a practical and rigorous introduction to applied biostatistics for health research using R. It covers the full analytical workflow: study questions, data types, descriptive statistics, visualization, inference, regression models, survival analysis, diagnostic accuracy, missing data, causal thinking, and reproducible reporting with Quarto.

The course emphasizes estimation, interpretation, reproducibility, and appropriate use of statistical methods in clinical research.

1.2 Target Audience

Clinicians, biomedical researchers, PhD students, public health professionals, and analysts. No advanced statistics background required.

1.3 Prerequisites

Basic computer literacy. No prior R experience required; R and RStudio setup is covered in Session 1.

1.4 Learning Objectives

By the end of BIOSTATR, participants will be able to:

1. formulate health research questions in statistical terms;
2. identify study designs and variable types;
3. organize reproducible analyses using R and Quarto;
4. summarize and visualize health data appropriately;
5. interpret confidence intervals, p-values, and effect estimates;
6. compare groups using suitable statistical tests;
7. fit and interpret linear and logistic regression models;
8. understand the basics of survival analysis and diagnostic accuracy;
9. recognize issues related to missing data and causal thinking;
10. produce a reproducible statistical report.

2 Detailed Schedule

Table 1: BIOSTATR detailed schedule

No.	Week	Date	Time	Topic	Main Content
1	1	07/09/2026 (Mon)	18h00- 20h00	Introduction to Biostatistics, Study Design & Reproducible R/Quarto Workflow	Role of biostatistics, study designs, data types, bias, confounding, R basics, RStudio projects, Quarto workflow
2	1	09/09/2026 (Wed)	18h00- 20h00	Descriptive Statistics & Data Visualization	Central tendency, dispersion, frequencies, missing values, Table 1, distribution plots, uncertainty visualization
3	1	11/09/2026 (Fri)	18h00- 20h00	Estimation, Statistical Inference & Group Comparisons	Sampling, confidence intervals, p-values, effect sizes, t-test, Wilcoxon, chi-square, Fisher, ANOVA, Kruskal-Wallis
4	1	13/09/2026 (Sun)	10h00- 12h00	Correlation and Linear Regression	Pearson/Spearman correlation, simple and multiple linear regression, diagnostics
5	2	14/09/2026 (Mon)	18h00- 20h00	Logistic Regression and GLMs	Binary outcomes, odds ratios, adjusted models, discrimination, calibration
6	2	16/09/2026 (Wed)	18h00- 20h00	Survival Analysis and Diagnostic Accuracy	Kaplan-Meier, censoring, Cox model, sensitivity, specificity, PPV, NPV, ROC curves
7	2	18/09/2026 (Fri)	18h00- 20h00	Missing Data, Causal Thinking, Sample Size & Reporting	MCAR/MAR/MNAR, multiple imputation, DAGs, confounding, power, reporting guidelines
8	2	20/09/2026 (Sun)	10h00- 12h00	Final Synthesis and Mini-Project	Complete analysis workflow, interpretation, Quarto report, participant presentations

3 Course Materials

- Slides and self-contained notebooks distributed before each session
- R scripts, datasets, and exercises (shared repository)
- Exercises reviewed and corrected during sessions

4 Assessment

Each course ends with a short reproducible Quarto report or mini-project, presented and discussed in Session 8.

5 About the Instructor

Imad EL BADISY, PhD, is a biostatistician working on statistical and machine learning methods for health data, with a focus on survival analysis. Full profile, publications, and R packages: elbadisyimad.com

6 Policies

Language: Teaching in French, with English for statistical and methodological terminology.

Software: R (≥ 4.4) and RStudio required; setup covered in Session 1.

Attendance: Full attendance at all 8 sessions is recommended given the cumulative structure.